Bell Pair Extraction and Bottlenecks in Grids and Related Networks

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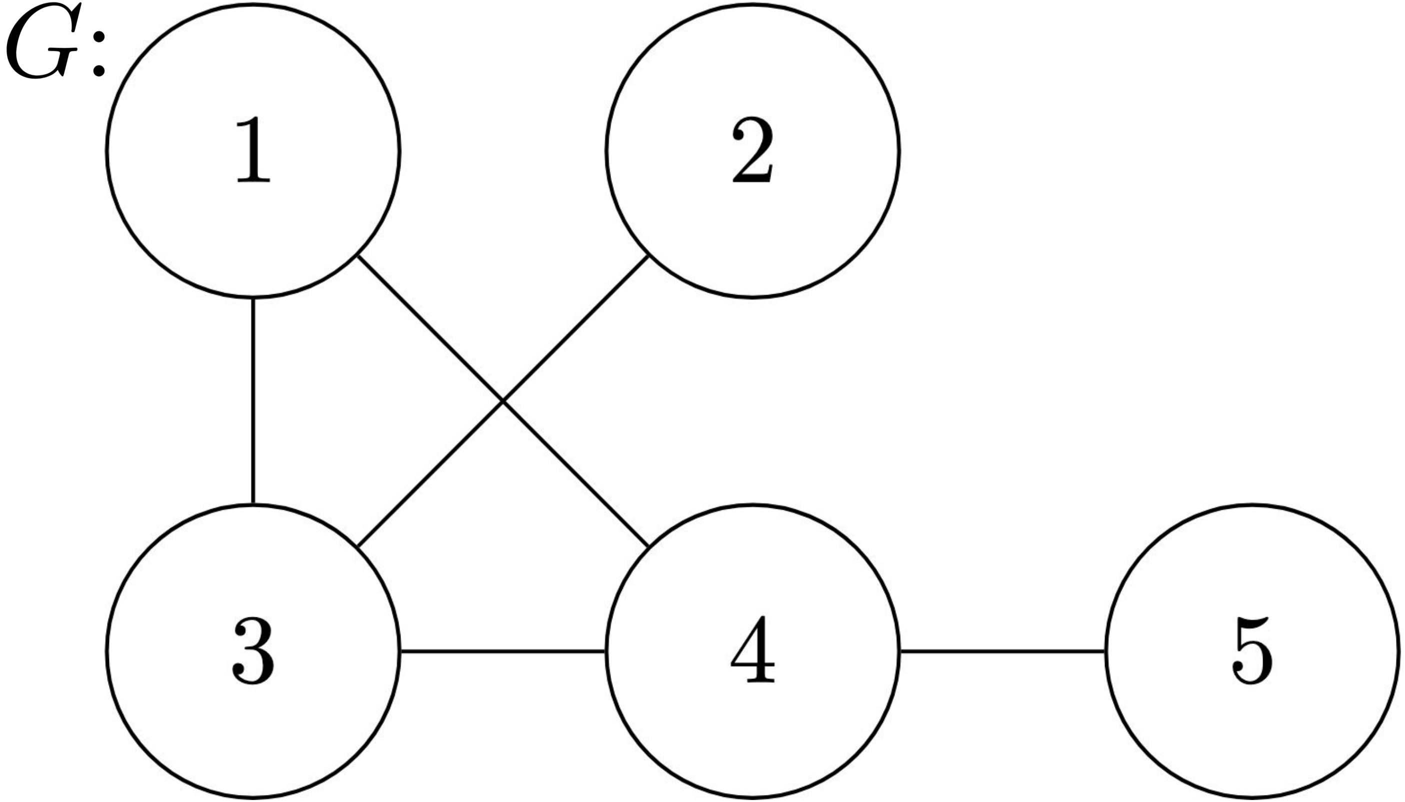
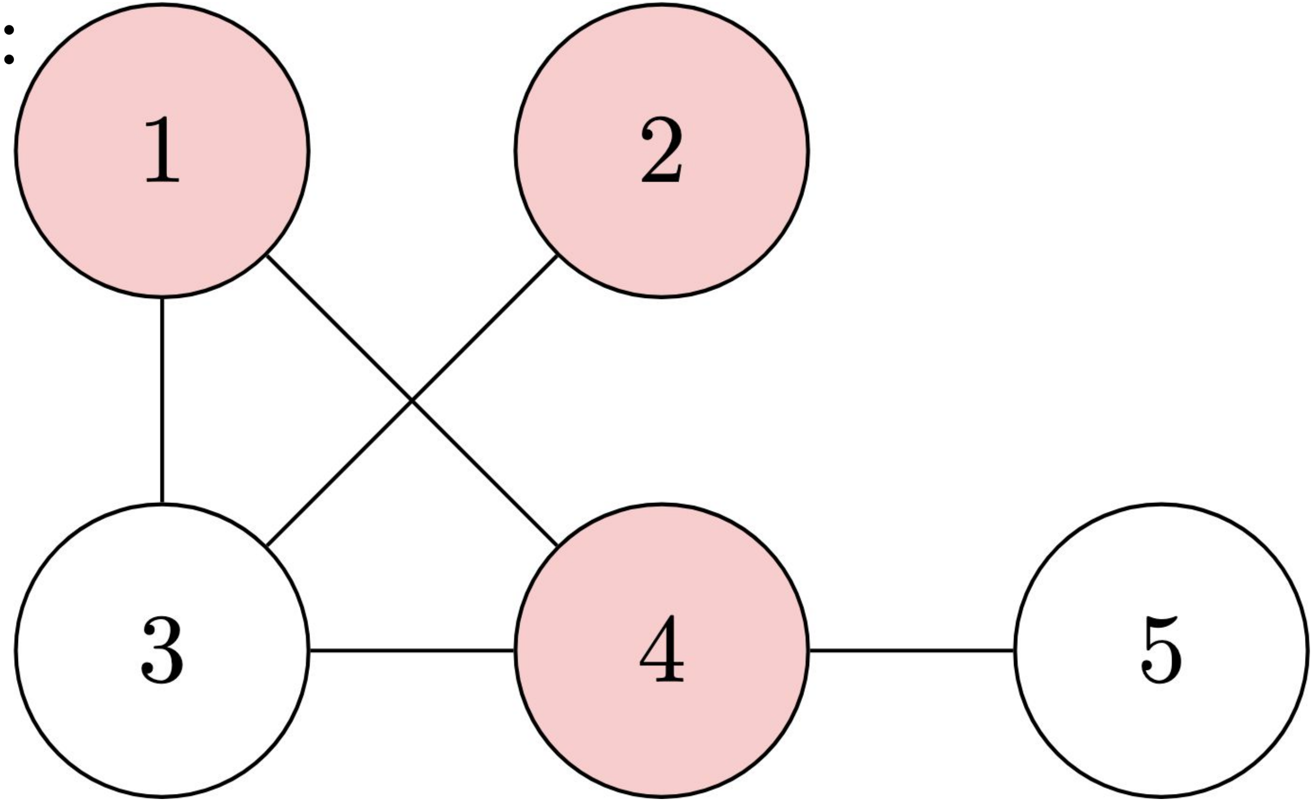
Definitions

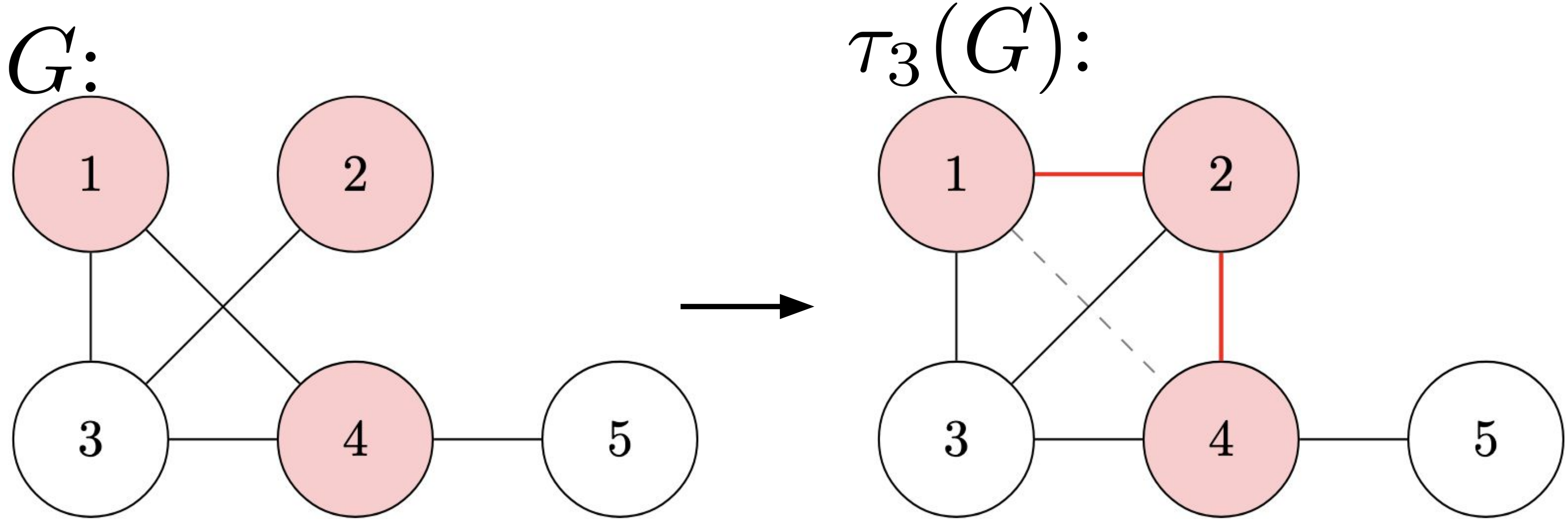


Definitions: Neighborhood

N_3:



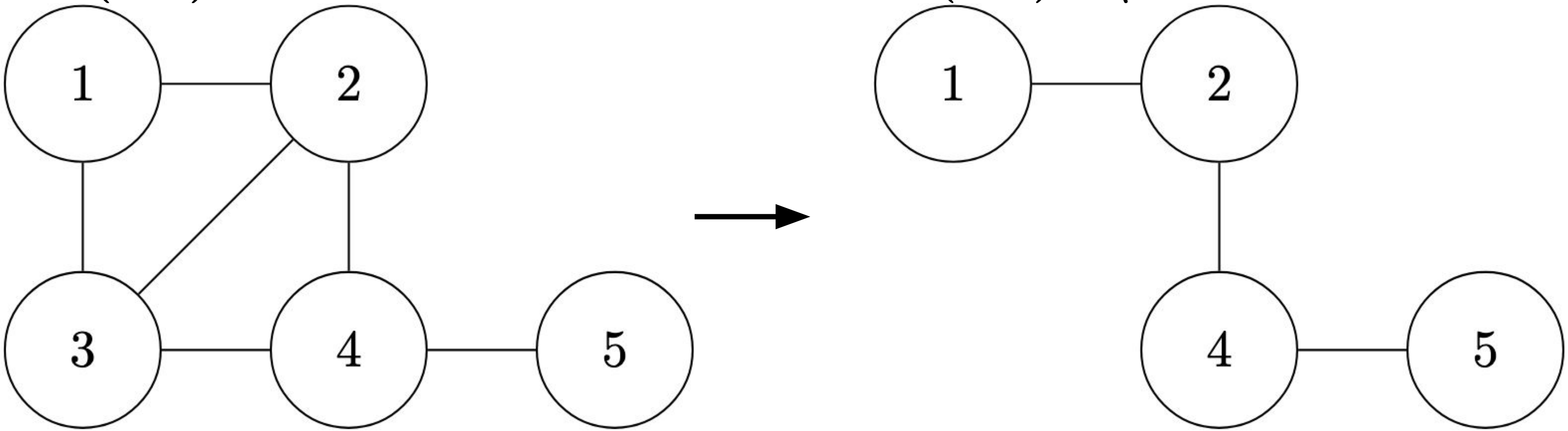
Definitions: Local Complementation (τ operation)



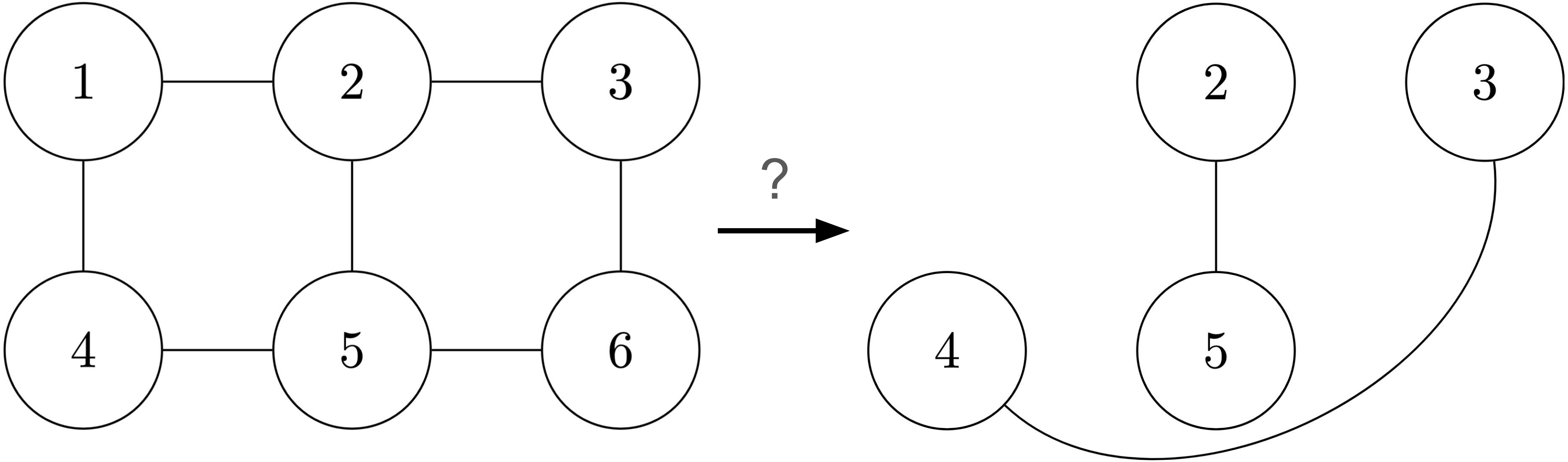
Neighborhood of vertex 3

Definitions: Vertex Deletion

$\tau_3(G)$: $\qquad\qquad$ $\tau_3(G) \setminus 3$:



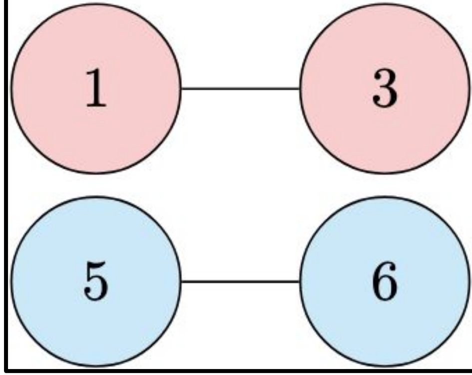
Definitions: Vertex-Minor Problem



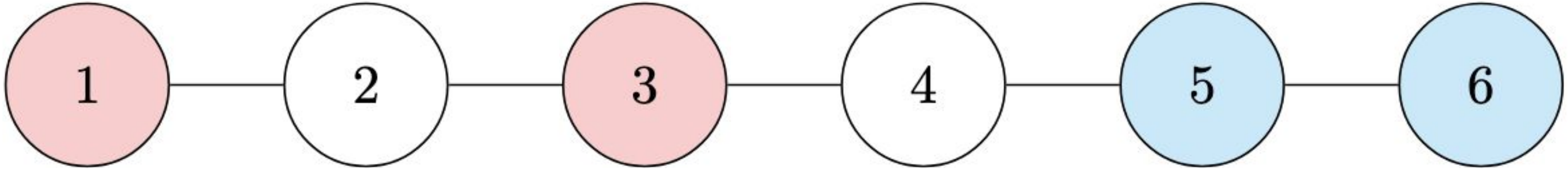
Operations: local complementation & vertex deletion

Example 1

Goal:
(Bell pair graph)



G:



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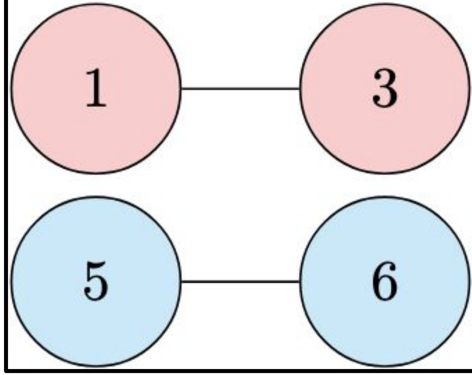
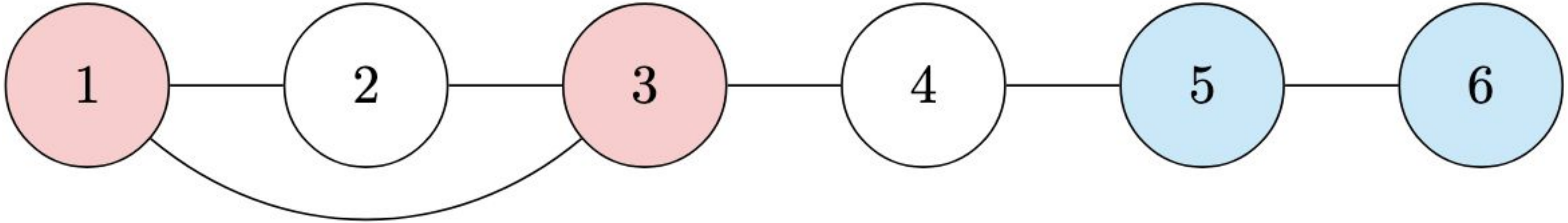


$\tau_2(G)$:



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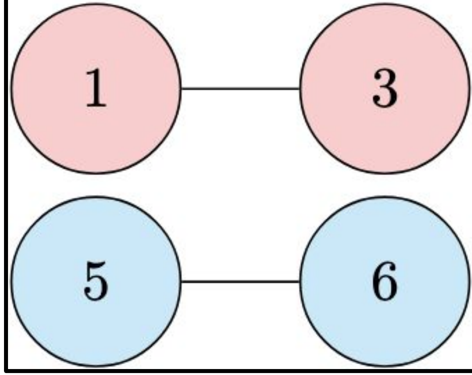
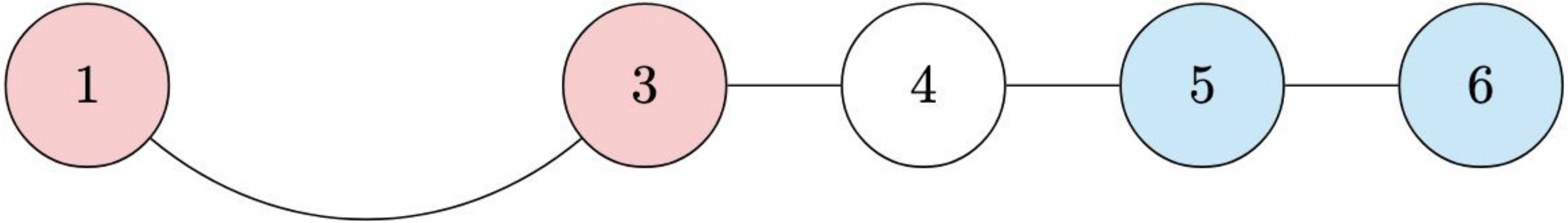


$\tau_2(G) \setminus 2$:



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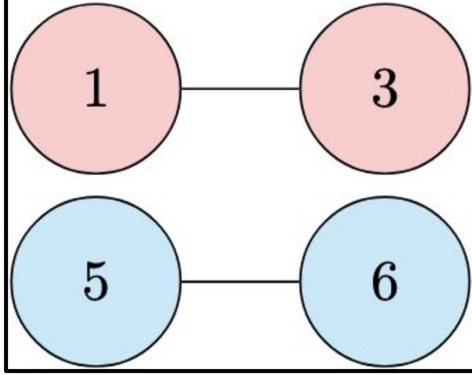
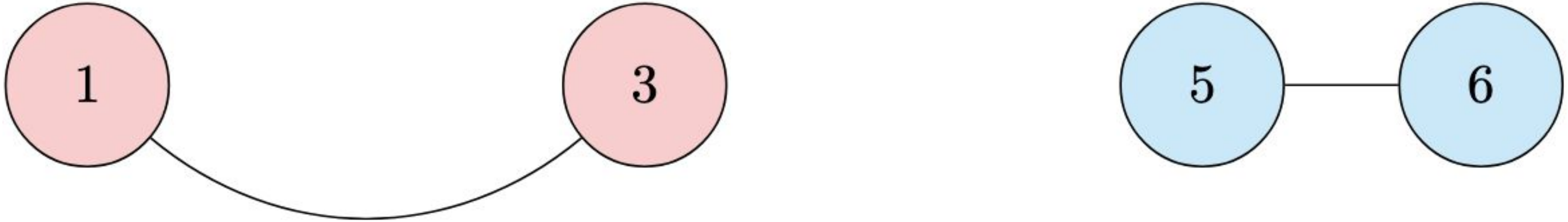
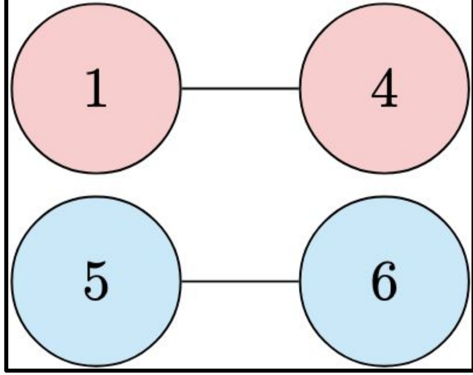


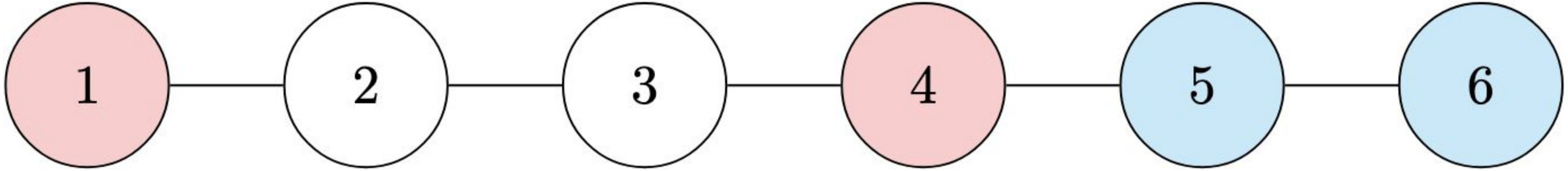
$(\tau_2(G) \setminus 2) \setminus 4$:



Example 2:

Goal:
(Bell pair graph)



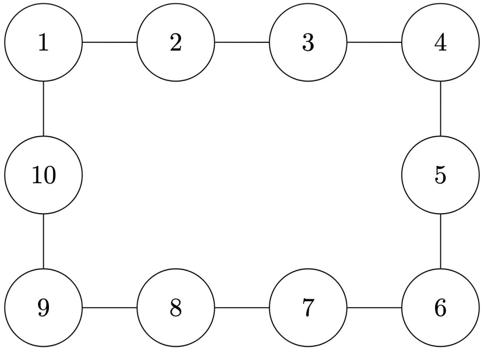


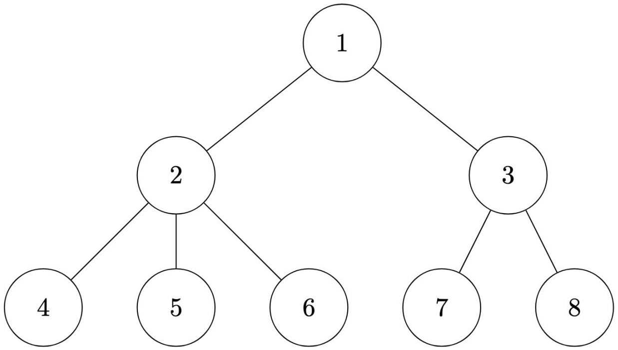
Quantum Computing/Science Motivation

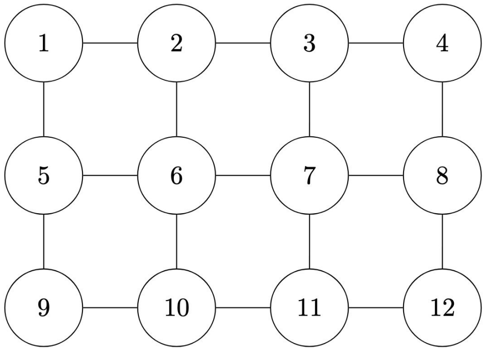
- Goal: Enable reliable quantum communication between distant computers/qubits
- Problem: Create special pairs of connected qubits (bell pairs) between chosen points in a quantum network
  - A quantum network can be thought of as connected quantum computers/qubits
- Solution: represent a quantum network as a graph where:
  - Vertices represent qubits
  - Edges represent quantum links (entanglement)
- We can only modify the network specified operations
  - Local Clifford operations
  - Pauli measurements
  - Classical Communication
- Ideally, we want to generate multiple Bell pairs simultaneously to support tasks like quantum teleportation
- Qubit-minor problem: Can the quantum network be transformed to create the desired pattern of connections?
  - Qubit-minor problem = vertex-minor problem (Dahlberg & Wehner 2018)

Past Results

- Some cases of ring & line graphs (Hahn, Eisert, & Pappa 2022)
- Some cases of grids (Mannalath & Pathak 2023)
- Solved ring, line, & tree graphs (Zhang 2025)





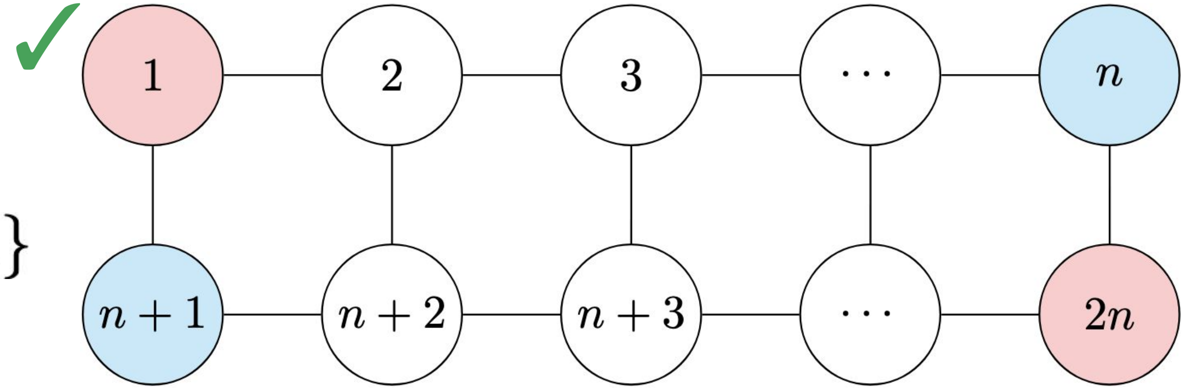


2xn Results - Mannalath & Pathak 2023

n odd:

- Can extract

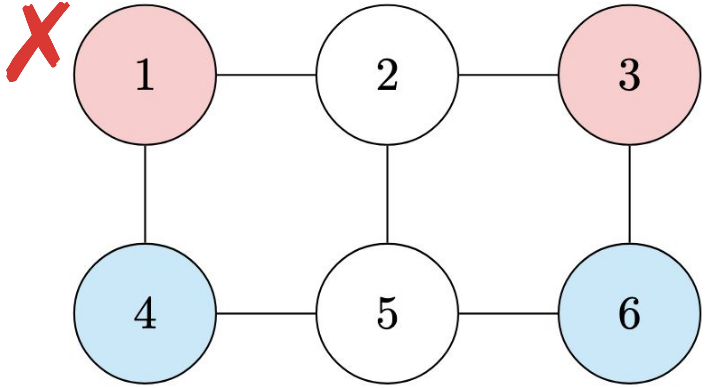
  $\{1, 2n\}$ & $\{n, n+1\}$



- Cannot extract

  $\{1, 3\}$ & $\{4, 6\}$

  when $n = 3$

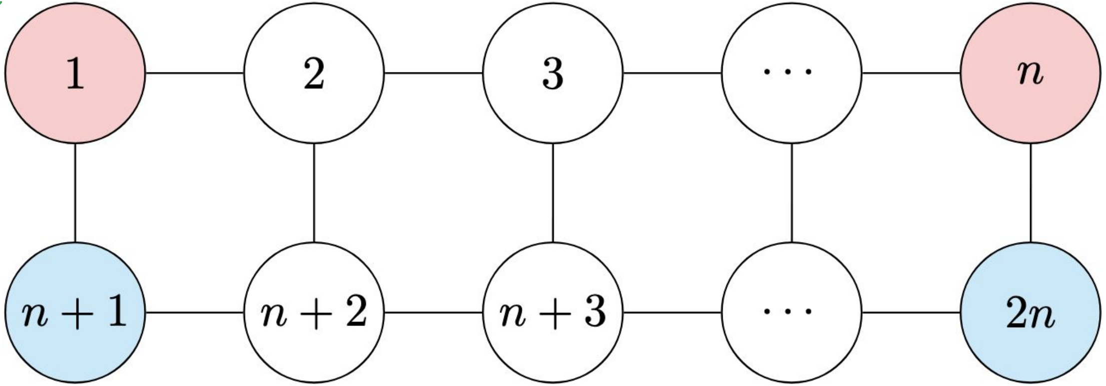


2xn Results - Mannalath & Pathak 2023

n **even:**

- Can extract

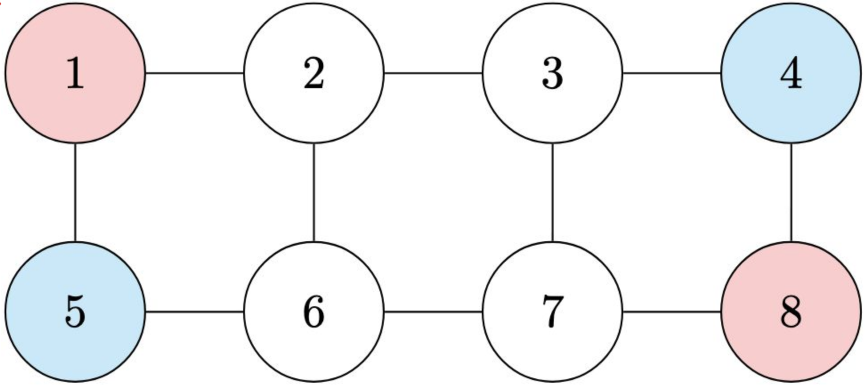
$\{1, n\}$ & $\{n, n+1\}$



- Cannot extract

$\{1, 8\}$ & $\{4, 5\}$

when $n = 4$



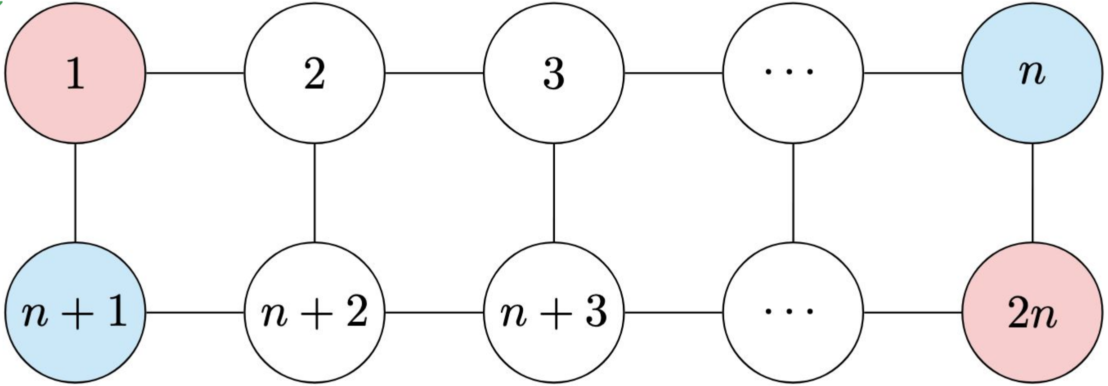
2xn Results - New

$n > 4$

- Can extract

  $\{1, 2n\}$ & $\{n, n+1\}$

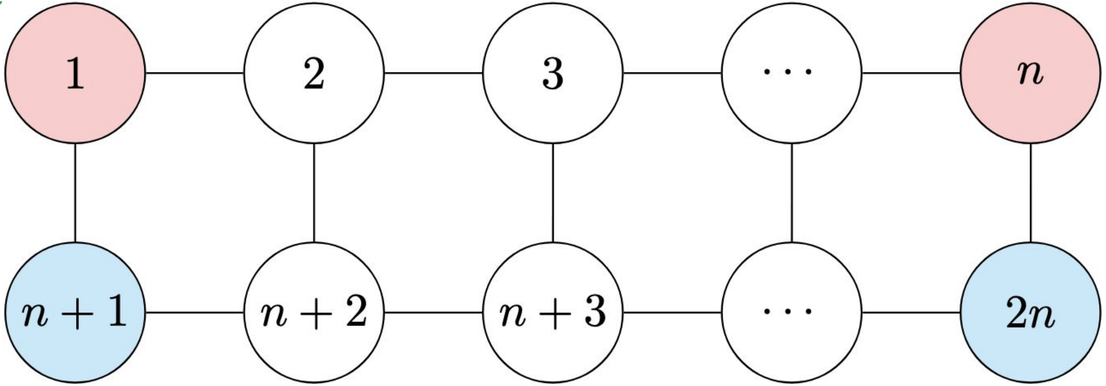
✓



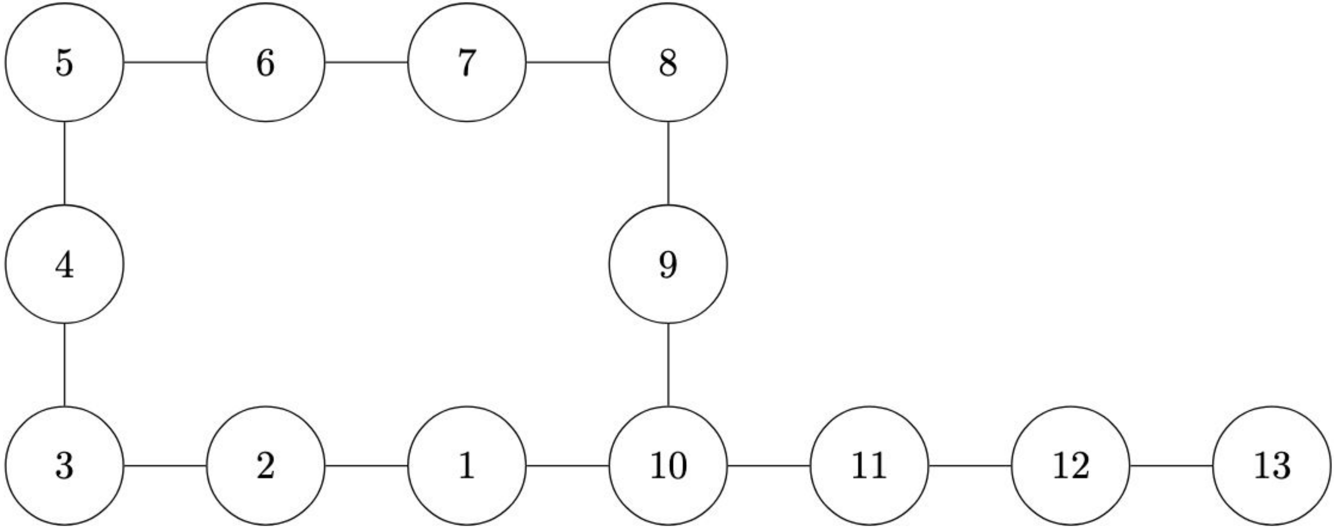
- Can extract

  $\{1, n\}$ & $\{n+1, 2n\}$

✓

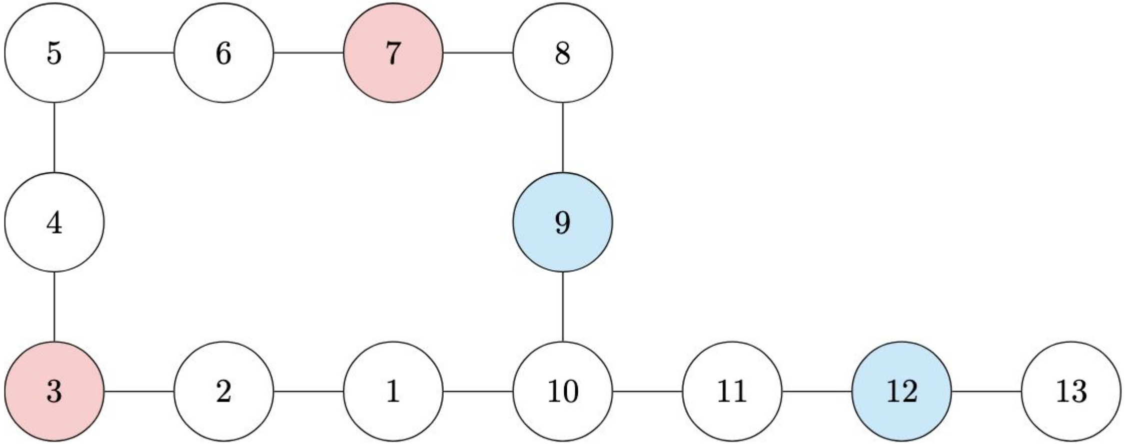


Tadpole Graphs



Tadpole Graphs

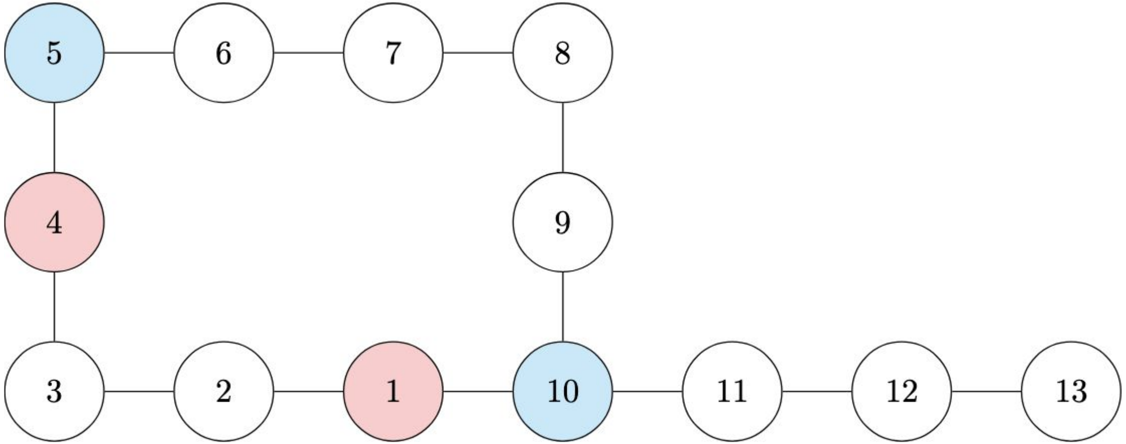
Can extract iff paths are non-adjacent or a special case is satisfied:



Tadpole Graphs

Can extract iff paths are non-adjacent or a special case is satisfied:

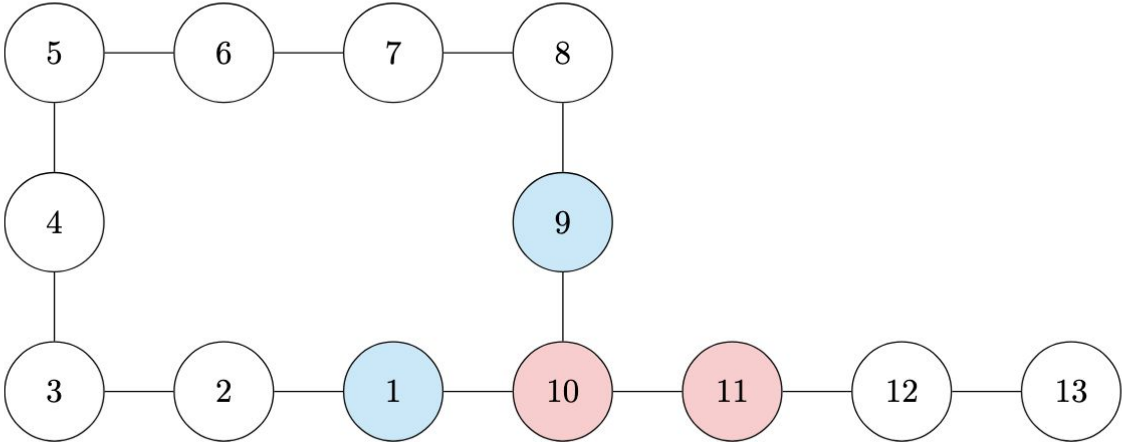
- all in ring, no 3 consecutive, paths do not cross



Tadpole Graphs

Can extract iff paths are non-adjacent or a special case is satisfied:

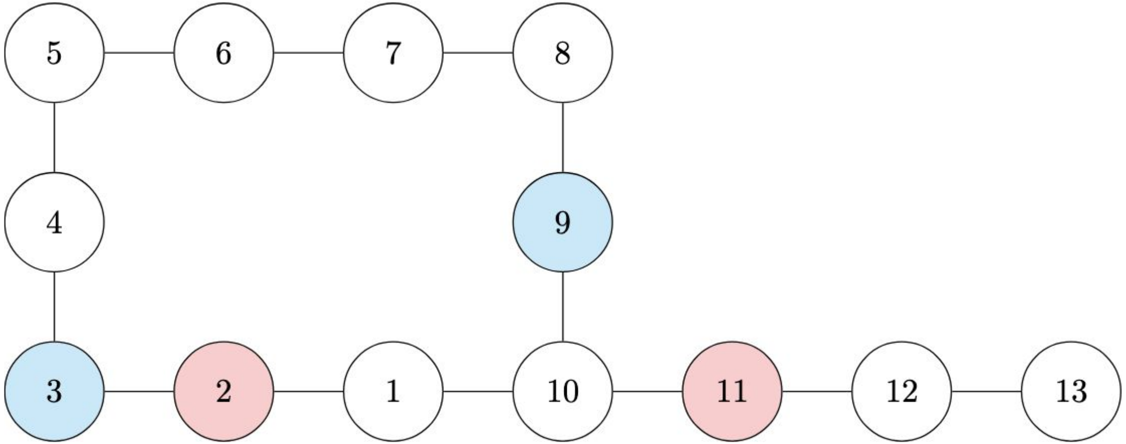
- exactly 3 vertices in ring including vertex 10, paths do not cross, and all 4 are not consecutive



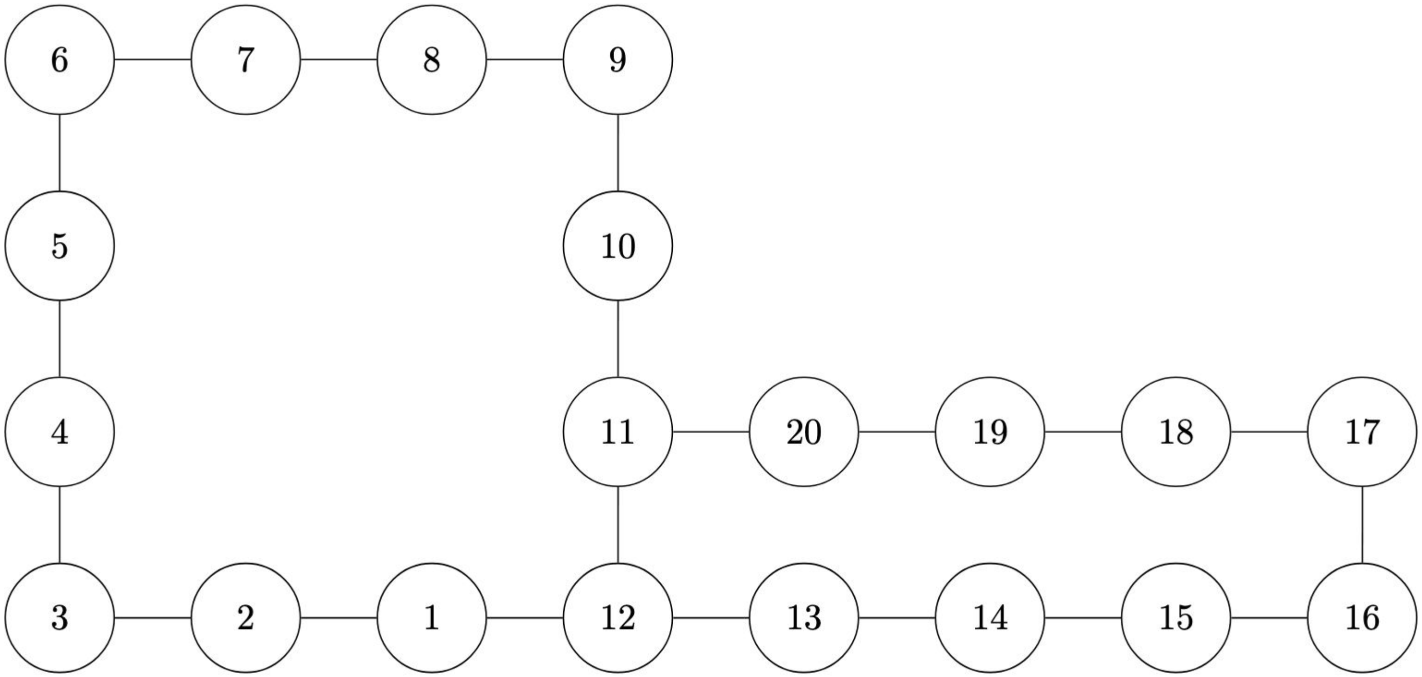
Tadpole Graphs

Can extract iff paths are non-adjacent or a special case is satisfied:

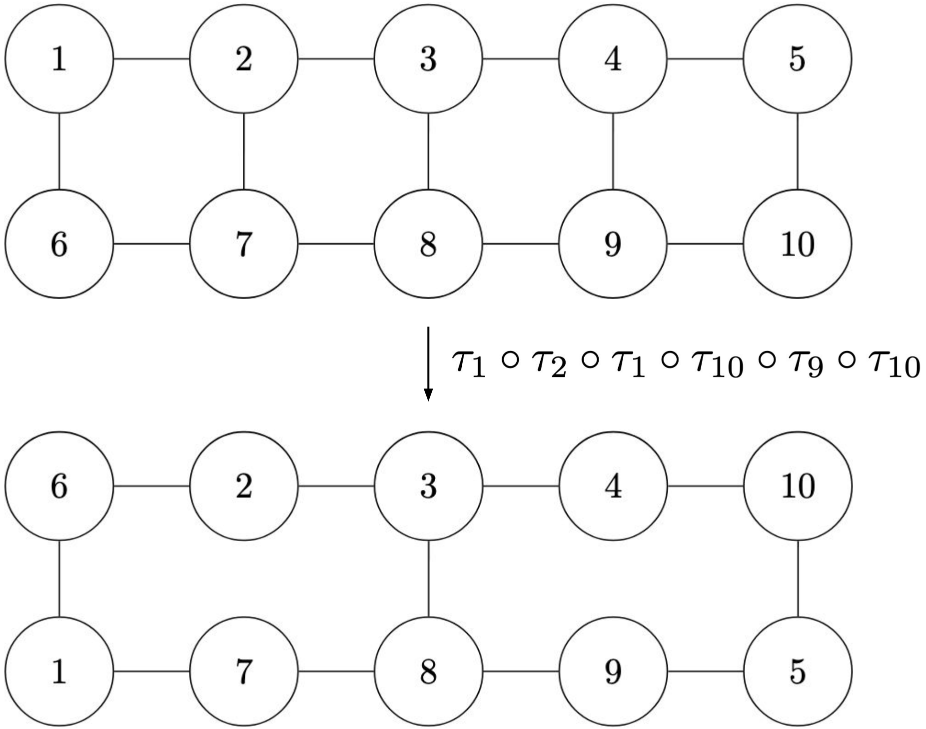
- exactly 3 vertices in ring excluding vertex 10, paths do not cross, and no 3 of the nodes in the ring plus node 10 are consecutive



Double Ring Graphs



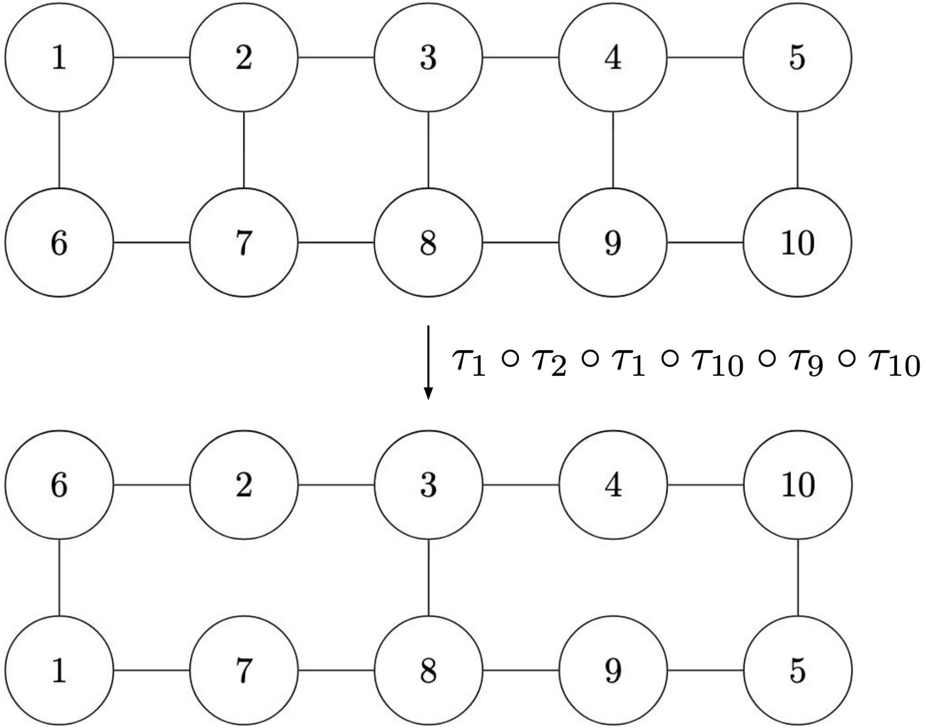
Corollary to 2x5 Grid



		b							
		1	2	4	5	6	7	9	10
a	1		✗	✓	✓	✗	✗	✓	✓
	2	✓		✓	✓	✗	✓	✗	✓
	4	✓	✓		✓	✓	✗	✓	✗
	5	✓	✓	✗		✓	✓	✗	✗
	6	✓	✗	✓	✓		✓	✓	✓
	7	✗	✗	✗	✓	✗		✓	✓
	9	✓	✗	✗	✗	✓	✓		✗
	10	✓	✓	✗	✓	✓	✓	✓	

(a) Extracting Bell pairs $\{a, 3\}, \{b, 8\}$

Corollary to 2x5 Grid



		b							
		1	2	4	5	6	7	9	10
a	1		✓	✗	✗	✓	✗	✗	✗
	2	✓		✗	✗	✗	✓	✗	✗
	4	✗	✗		✓	✗	✗	✓	✗
	5	✗	✗	✓		✗	✗	✗	✓
	6	✓	✗	✗	✗		✓	✗	✗
	7	✗	✓	✗	✗	✓		✗	✗
	9	✗	✗	✓	✗	✗	✗		✓
	10	✗	✗	✗	✓	✗	✗	✓	

(b) Extracting Bell pairs $\{a, b\}, \{3, 8\}$

3xn Results

- We looked at the border cases of a 3xn grid
- These seemed to be the most useful since most interior case can be transformed into border cases
- We were able to solve all of the “side” border cases

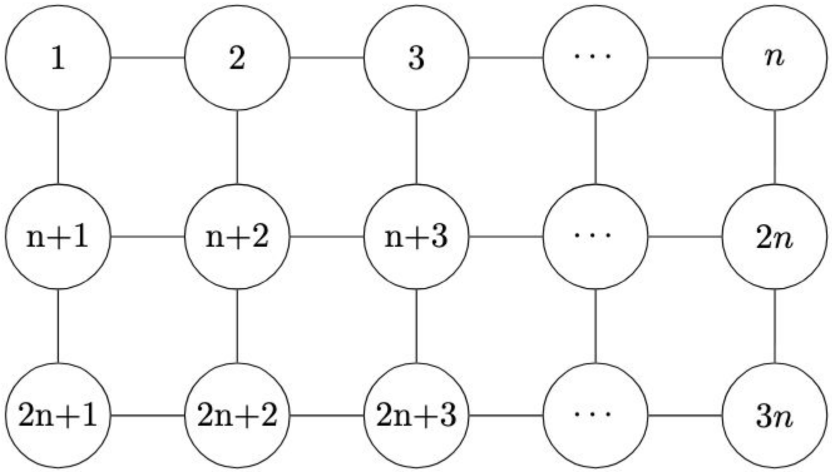
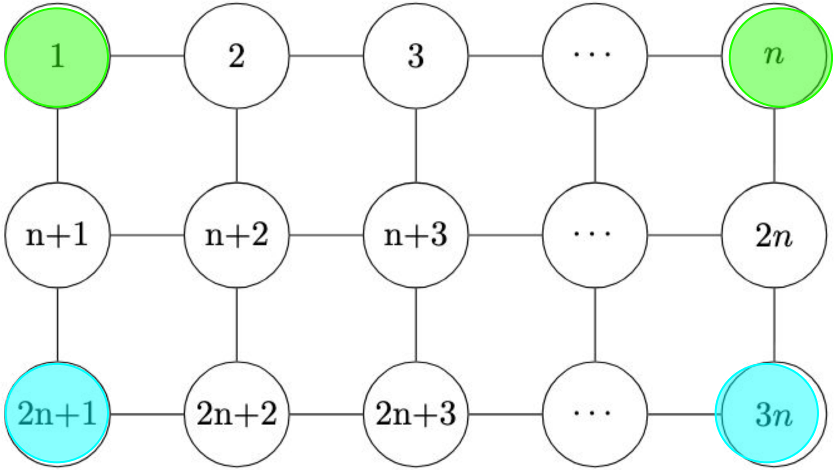


Figure 1: $3 \times n$ Grid

3xn Results

We can simultaneously extract all non-crossing side bell pairs:

- {1,n} & {2n+1, 2n} for n>1
- {1,n} & {n+1, 2n} for n>3
- {1,n} & {2n+1, 2n} for n>3
- {1, b} & {n+1, 2n+1}, where b = {2n, 3n} for n>2, and b= {n} for n>3



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*many of these results can be applied to further similar combinations of the graph based off of symmetry.

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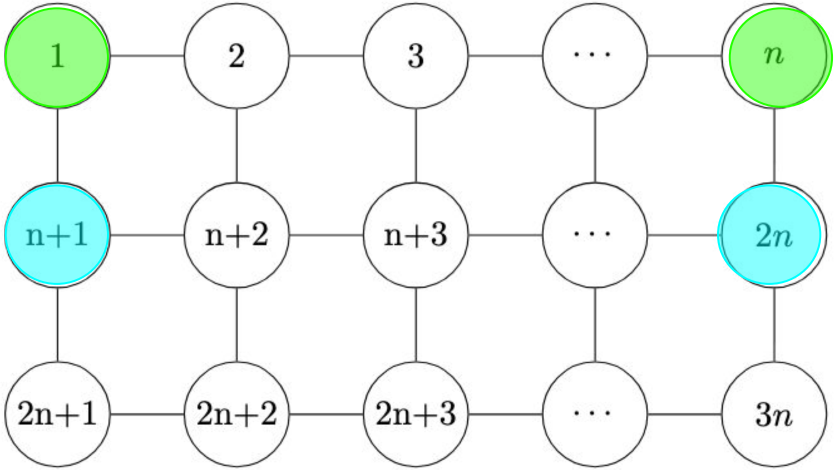

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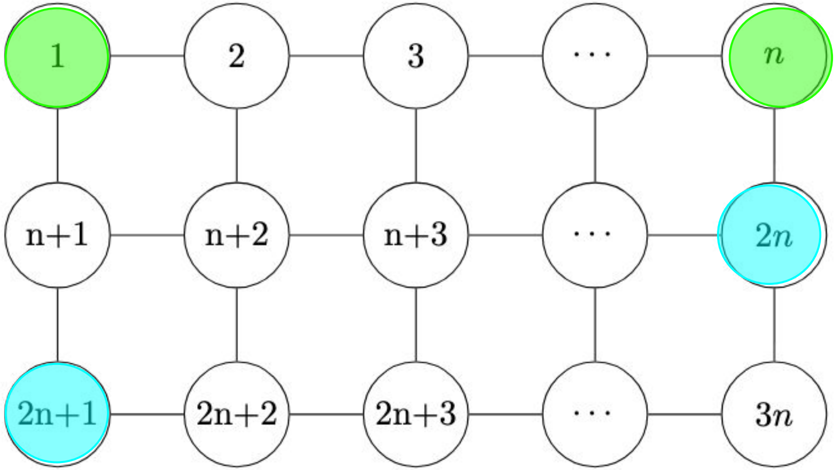

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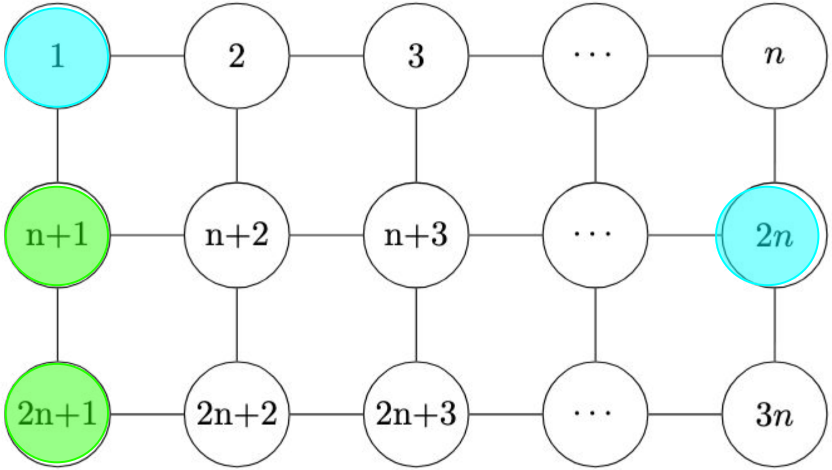

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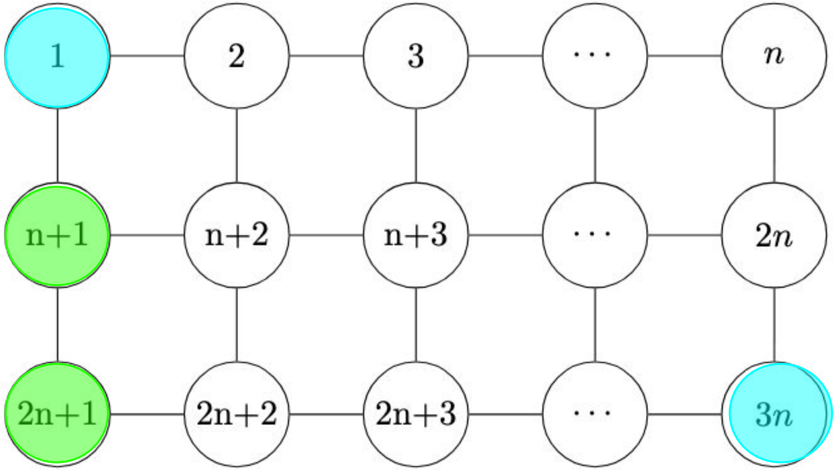

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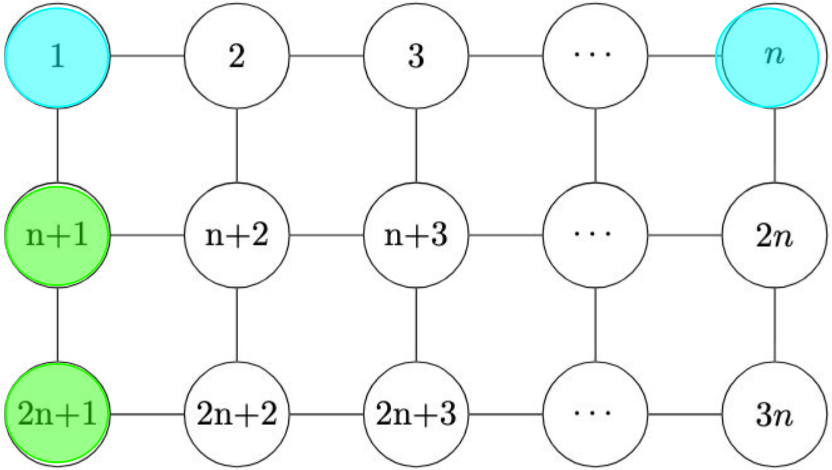

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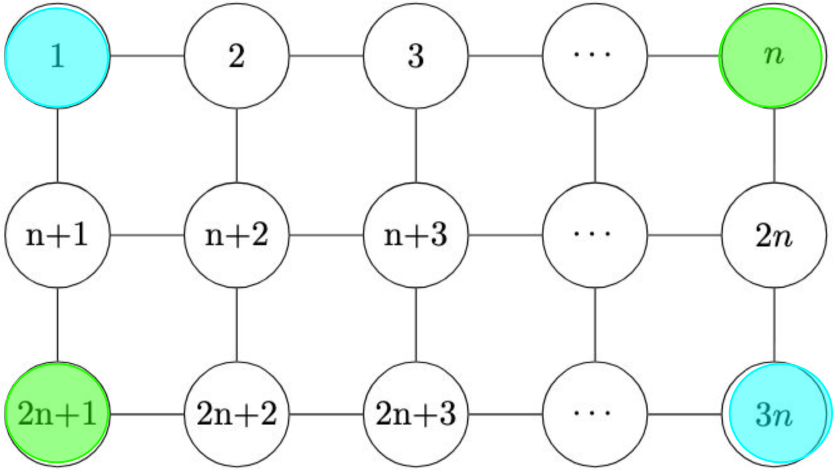
3xn Results

We can simultaneously extract all crossing side bell pairs:

- {1,3n} & {2n+1, n} for n>2
- {1,2n} & {n+1, n} for n=3 & n>4
- {1,2n} & {2n+1, n} for n>2
- {1, 3n} & {n+1, 2n} for n>5
- {1,2n+1} & {n+1, b}, where b = {n, 3n} for n>2, and b ={2n} for n>4



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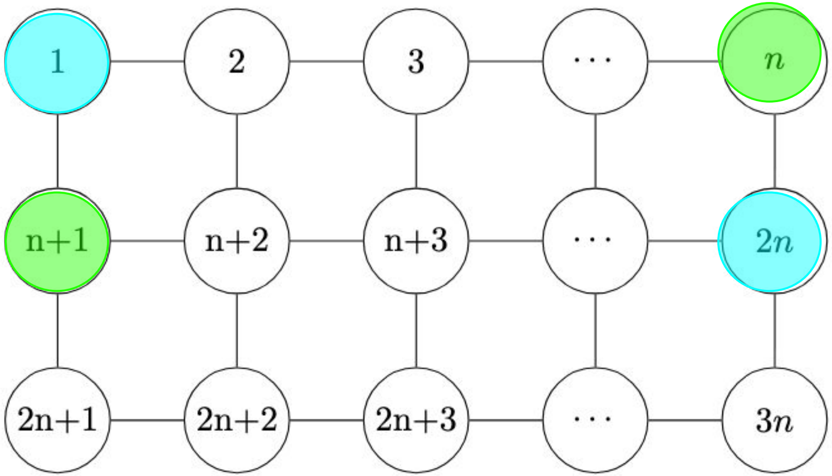

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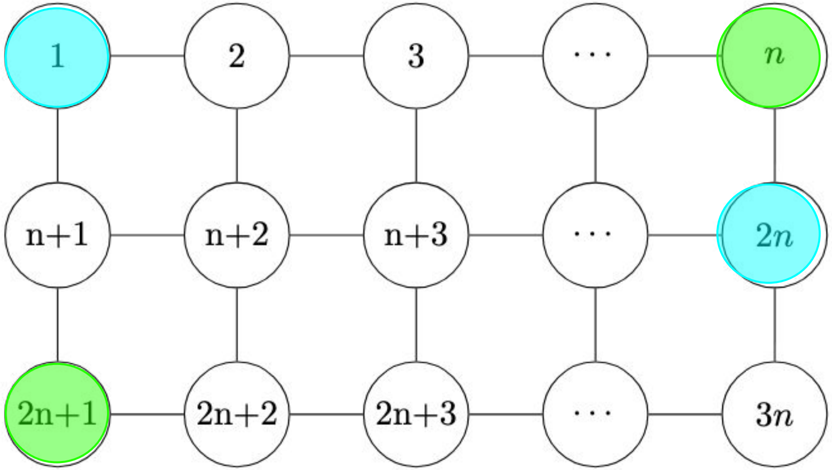

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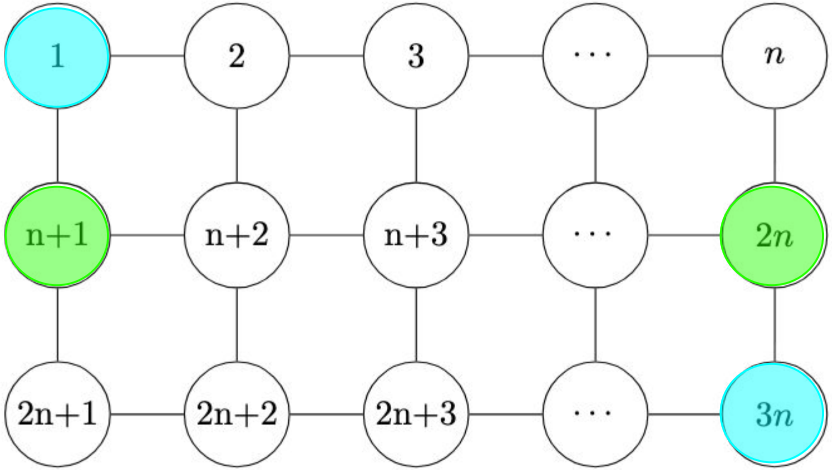

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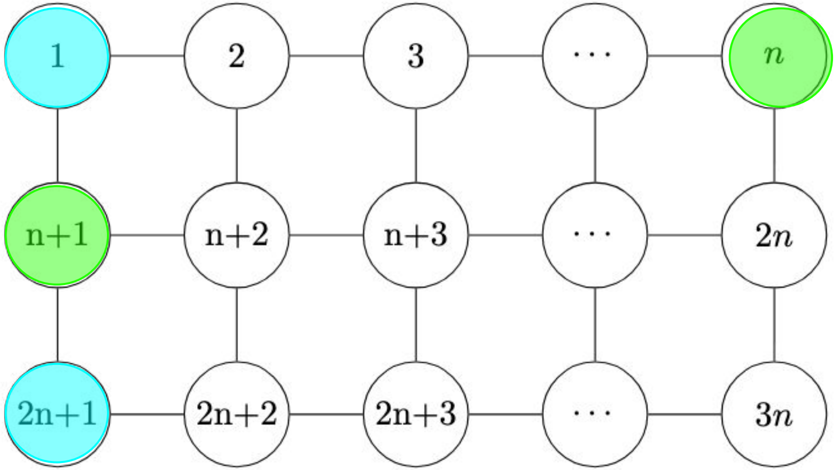

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- {1,2n+1} & {n+1, b}, where b = {n, 3n} for n>2, and b ={2n} for n>4

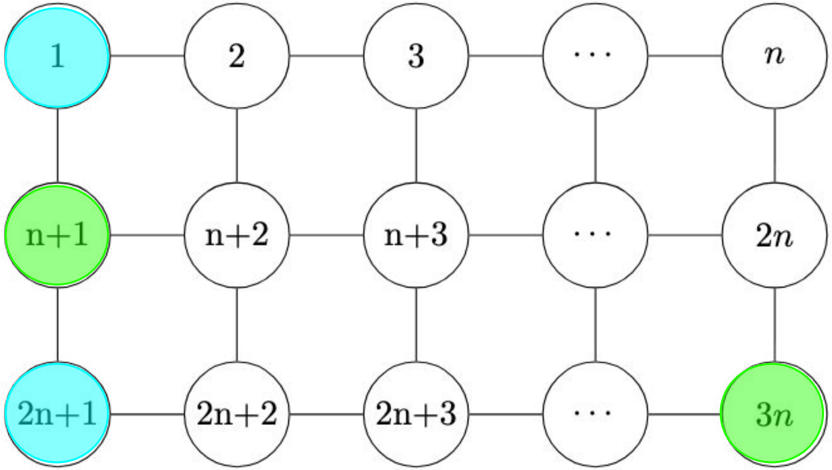


Figure 1: $3 \times n$ Grid

*many of these results can be applied to further similar combinations of the graph based off of symmetry.

3xn Results

We can simultaneously extract all crossing side bell pairs:

- {1,3n} & {2n+1, n} for n>2
- {1,2n} & {n+1, n} for n=3 & n>4
- {1,2n} & {2n+1, n} for n>2
- {1, 3n} & {n+1, 2n} for n>5
- {1,2n+1} & {n+1, b}, where b = {n, 3n} for n>2, and b ={2n} for n>4

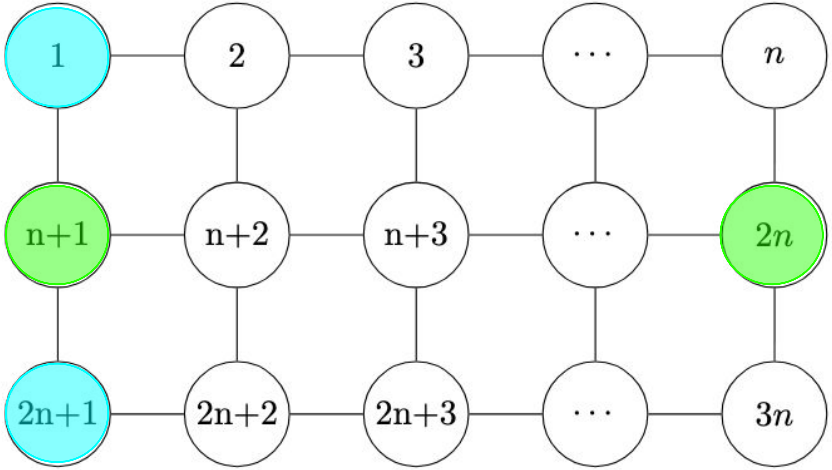
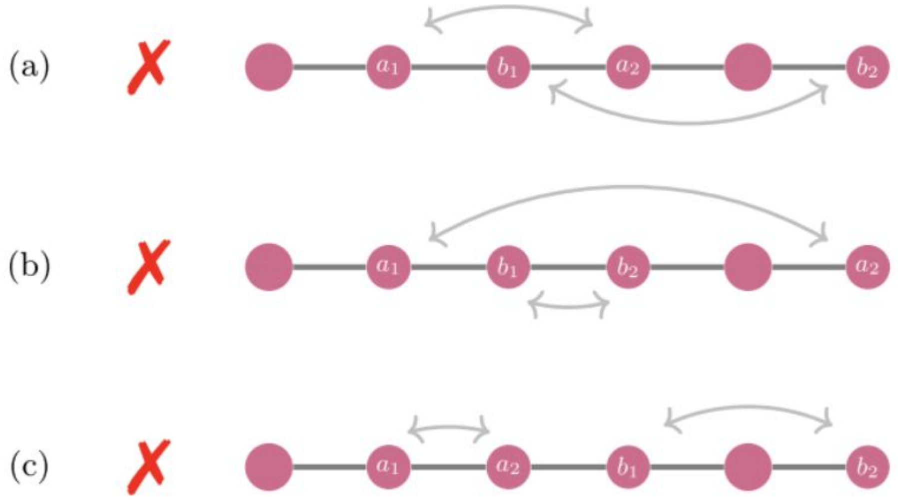


Figure 1: $3 \times n$ Grid

*many of these results can be applied to further similar combinations of the graph based off of symmetry.

Line Graph Bottlenecks

For a line graph, it is impossible to simultaneously extract:



1. “Overlapping” Bell Pairs
2. “Nested” Bell Pairs
3. “Neighboring” Bell pairs

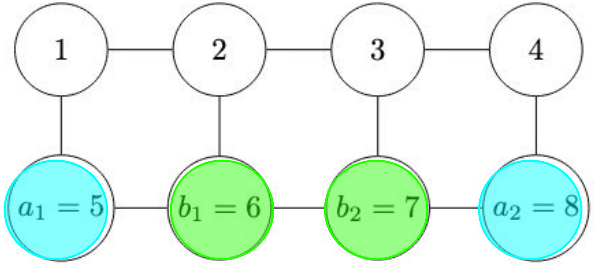
We wanted to see if it was possible to overcome such bottlenecks, when the network is extended to a grid.

Figure taken from, (Zhang 2025)

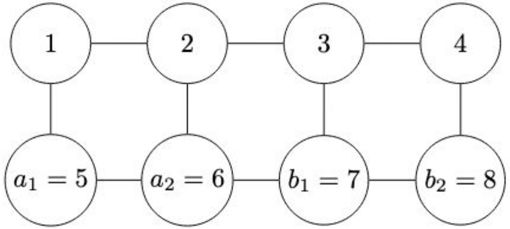
Overcoming Line Graph Bottlenecks

When extended to a 2xn Grid, we found that the following configurations were now extractable:

1. Nested Bell Pairs for n >3
   a. The proof follows by induction using a base case of n=4



2. Adjacent Bell Pairs for n>3
   a. Proof follows by establishing a protocol to split the graph into two sides

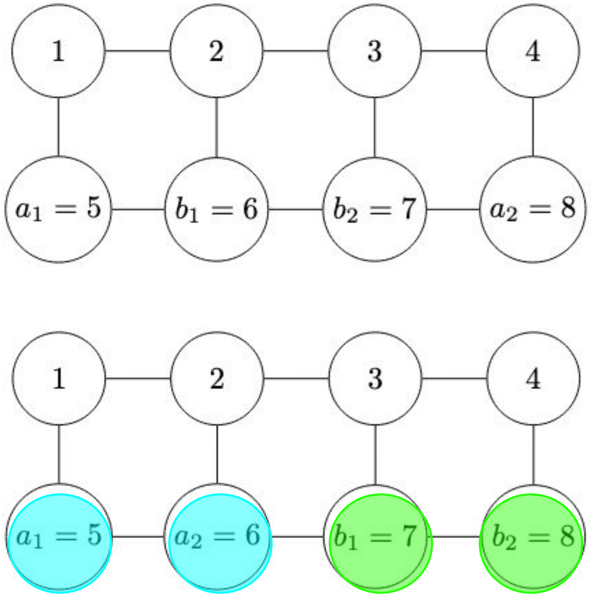


Overcoming Line Graph Bottlenecks

When extended to a 2xn Grid, we found that the following configurations were now extractable:

1. Nested Bell Pairs for n >3
   a. The proof follows by induction using a BC of n=4

2. Adjacent Bell Pairs for n>3
   a. Proof follows by establishing a protocol to split the graph into two sides



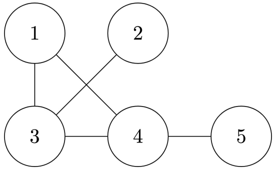
Overcoming Line Graph Bottlenecks

- We are still unsure if the “overlapping” Bell Pair bottleneck can be relieved by extending to a 2xn grid
- We suspect that using existing protocols we’ve created, some cases of overlapping bell pairs can be resolved in a 3xn grid
- More work will need to be done though to prove such solutions exist

Properties invariant under local complementation

- Cut-rank
  - "Amount" of connectedness
  - Monotonic with respect to G
    - If G' is vertex minor to G, then $\mathrm{cutrk}_{G'}(S) \leq \mathrm{cutrk}_{G}(S)$

G:



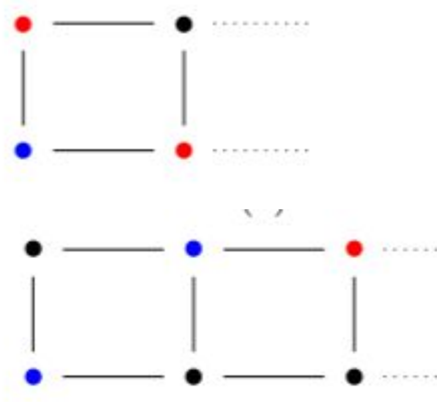
$$\mathrm{cutrk}_G(\{1,2,4\}) = \mathrm{rank}\left(\begin{array}{cc} & \begin{array}{cc} v_3 & v_5 \end{array} \\ \begin{array}{c} v_1 \\ v_2 \\ v_4 \end{array} & \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} \end{array} \right) = 2$$

  - To show H is not vertex minor to G, we need to find a set whose cut-rank on H is > than on G
  - Unfortunately, every cut-rank on a m×n graph with ≥ 2 in each collection has cut-rank ≥ 2
  - For any 4 vertices a, b, c, & d, there is a vertex minor G' such that $\mathrm{cutrk}_{G'}(\{a,c\}) = 2$
  - Not possible to distinguish between them.

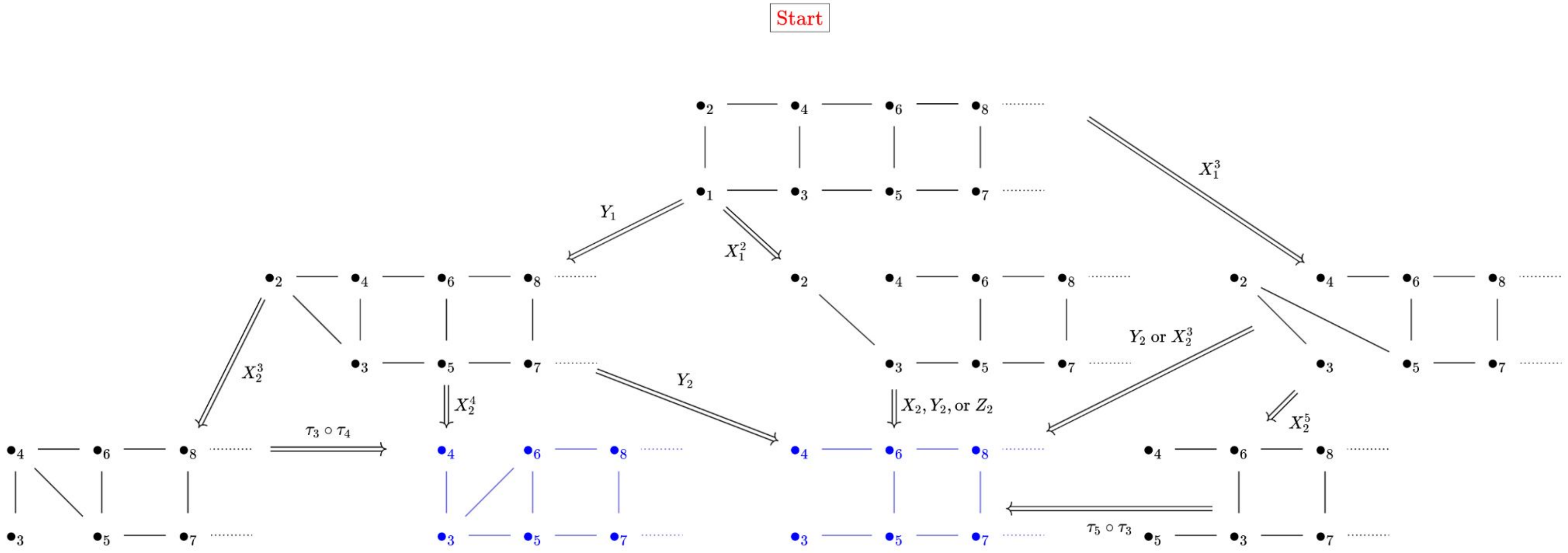
Properties invariant under local complementation

- Local sets
  - X is local set of G if X = S ∪ Odd(S)
  - X a set of vertices of G', if G' vertix minor to G and X local set of G,

    then S is a local set of G'
  - Contrapositive: S local set of G but not of G', then G' not vertex minor
  - Bell pair graph has no local sets of size 3
  - Grids have exactly the above local sets of size 3



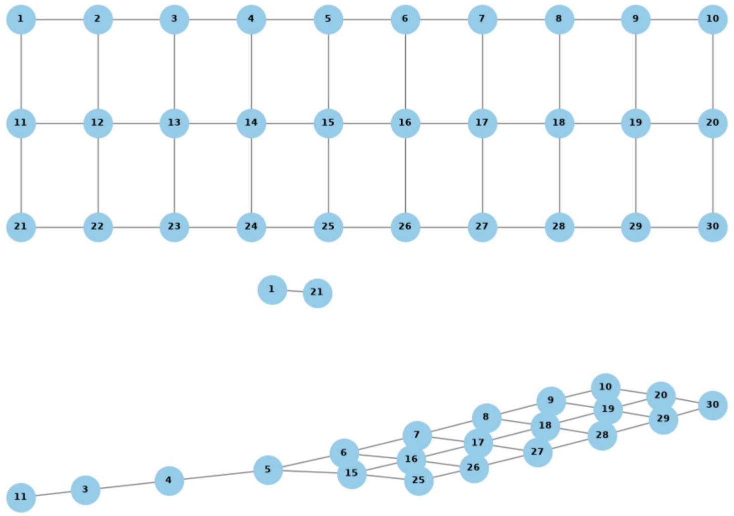
Adding a column



Summary of Progress Made

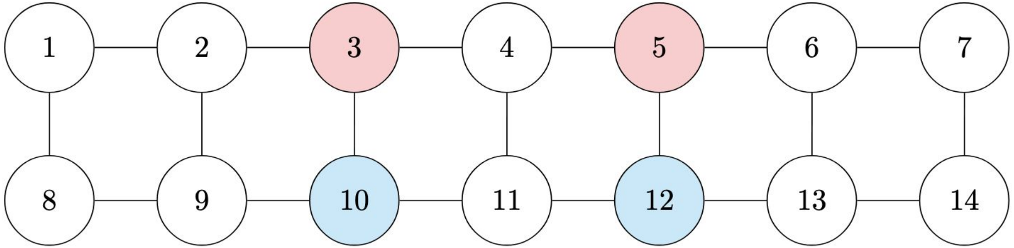
- Programmed code for visualizing graph operations
- Found constructions to extract Bell pairs in many orientations for 2xn & 3xn grids
- Found bottlenecks for extracting Bell pairs for 2xn & 3xn grids
- Solved tadpole graphs
- Solved some cases in double ring graphs

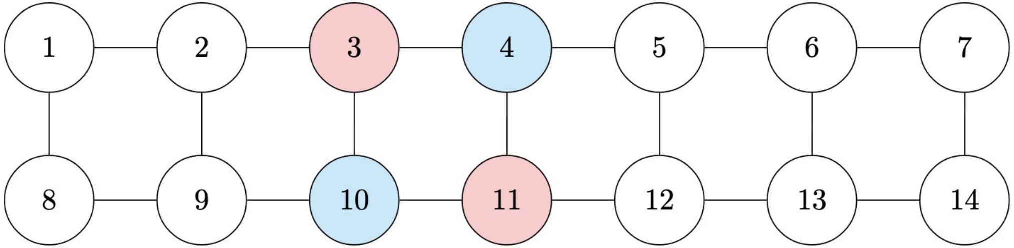
```
draw_graph(G_int, pos_int, (10,3))
##extract corners
G3 = extract_corners(G_int, cols, show_graph= False, gridlock = True, plot_size= (10,3), position_int= pos_int)
draw_graph(G3, plot_size =(10,3))
```



Open Questions

- Large grids where Bell pair nodes are close together
- Other cases in double ring graphs





Thank you!

- Derek for mentoring us
- Dr. Washington for running the REU program

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